

ASWIN A

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DOB:16-03-2006 | Nationality: Indian | Location: Chennai, India

Professional Summary

Electronics and Communication Engineering student with hands-on experience building software, IoT systems, data pipelines, REST APIs, and cloud-connected applications using Python, SQL, Docker, Git, MQTT, Kafka, and Azure. Experienced in developing end-to-end solutions involving data ingestion, backend processing, databases, machine learning, and real-time analytics. Strong problem-solving skills with a focus on AI-assisted development, scalable software components, testing, and production-oriented engineering.

Education

B. Tech - Electronic and Communication Engineering
S.R.M Institute of Science and Technology, Ramapuram, Chennai
CGPA: 8.76/10

Aug 2023 – May 2027

Technical Experience

1) Embedded System Developer – Virtual Internship [LINK](#)

Oct 2024 – Dec 2024

AICTE EduSkills, in association with Microchip

- Completed a 10-week industry-supported focused on embedded systems development and microcontroller-based applications.
- Strengthened structured problem-solving, development workflows, and practical engineering skills through hands-on technical exercises.

2) Data Science Intern – SQROCK IT Solution

Aug 2026 – Sep 2026

- Working on a structured project-based internship covering machine learning, forecasting, and NLP applications through real-world data science tasks.
- Strengthening model building, data preprocessing, and evaluation skills through hands-on project development and practical exercises.

Technical Projects

1) IoT Factory Monitoring & Streaming Analytics [LINK](#)

- Developed a real-time IoT data pipeline to stream factory sensor data from ESP32 (Wokwi) using MQTT and Apache Kafka.
- Implemented an MQTT-based data ingestion architecture using Wokwi, EMQX, Python, and PostgreSQL.
- Automated sensor data processing and storage through an ETL pipeline for scalable real-time analytics.
- Built a live Power BI dashboard to monitor machine performance, environmental conditions, and operational KPIs.

Tech Stack: Python, Apache Kafka, Spark, MQTT, Docker, PostgreSQL, SQL, Power BI, ESP32 (Wokwi)

2) Earthquake Auto-Shutdown System – IoT & Edge ML [LINK](#)

- Applied a real-time seismic detection system achieving ~97% vibration classification accuracy using Edge Impulse ML models.
- Improved an end-to-end IoT pipeline transmitting sensor data and emergency alerts to the cloud via Azure IoT Hub (MQTT).
- Programmed relay-based automated safety shutdown mechanisms to immediately power down machinery during detected seismic events.
- Demonstrated practical integration of embedded systems, machine learning, and cloud communication in safety-critical scenarios.

Tech Stack: Embedded C/C++, ESP32, MPU6050, Edge Impulse, Machine Learning, IoT, MQTT, Azure IoT Hub

3) Global Cost of Living Analysis using Price Level Index [LINK](#)

- Analyzed 6,800+ World Bank records to identify regional patterns, anomalies, and purchasing-power trends across countries.
- Performed data cleaning, validation, feature engineering, outlier detection, and statistical analysis using Python and Pandas.
- Investigated unusual observations and country-level variations to identify meaningful patterns from large datasets.
- Built interactive Power BI dashboards to communicate regional trends, KPIs, and analytical findings through maps and visualizations.

Tech Stack: Python, Pandas, Scikit-learn, TensorFlow, Power BI, Matplotlib

Technical Skills

Programming & Databases: Python, C++, Bash, PostgreSQL, SQLite

Analytics & Engineering: Power BI, Excel, EDA, Statistical Analysis, Data Cleaning & Visualization, Linux, Apache Kafka, MQTT

Tools & Frameworks: Pandas, NumPy, Flask, REST APIs, Database Design, Git, GitHub, Postman, VS Code, Docker

Soft Skills: Problem Solving, Teamwork, Time Management, Self-Directed Learning

Certifications & Activities

- Python for Data Science [LINK](#)
- GenAI Powered Data Analytics [LINK](#)
- Data Science & Analytics [LINK](#)
- Machine Learning Workshop [LINK](#)